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1. Introduction

1.1 Background

A key issue in energy economics, smart grid management, and sustainability planning includes predicting usage of electricity. Accurate forecasting facilitates efficient load balancing, infrastructure investment, and the development of policies targeted at lowering emissions and energy waste. Understanding household energy usage trends are increasing as urbanization and electrification have increased.

Time series analysis provides a structured statistical framework to study historical electricity consumption data and to produce reliable forecasts. Predictive accuracy can be increased, and underlying consumption behavior can be interpreted by breaking down a series into trend, seasonal, and residual components.

1.2 Data Description

The Household Power Consumption dataset, describing power usage from a single home in Sceaux, France, was used in this analysis. Between December 2006 and November 2010, measurements were made at a one-minute sampling rate, contains 2,075,259 measurements. There are different measurements from different electrical quantities, but this analysis concentrates on Global Active Power, which is the total amount of power consumed by households expressed in kilowatts. Aggregation by month was required to enable Time series analysis because of the original data's extremely high frequency at minute level.

A final dataset with 48 complete monthly observations from December 2006 to November 2010 was produced by aggregating the data to monthly average consumption.

1.3 Aim and Objectives

The main aim of this study is to analyze and forecast household electricity consumption using time series methods.

The specific objectives are:

To clean and preprocess the electricity consumption data by:

- Missing values in the Global_Active_Power variable are estimated by interpolation of previous and posterior observations to ensure a complete and consistent time series.
- To explore the statistical properties of the time series using summary statistics.
- To identify seasonal and trend patterns to help choose the right model.
- To compare Classical time series methods and ARIMA-based models for forecasting electricity consumption.
- To forecast and evaluate 12-month electricity consumption forecasts

2. Literature Review

Electricity demand forecasting has been extensively studied using both Classical and Time Series approach. Classical time series models, including Simple Exponential Smoothing, Holt's, and Holt-Winters models, are applied when data exhibit clear trend and seasonal patterns. These models are appropriate for operational forecasting since they are computationally effective and simple to understand.

The ARIMA modelling framework, which allows for flexible modelling of stationarity condition and autocorrelation structures. Seasonal ARIMA (SARIMA) models are used to capture periodic patterns observed in the data.

Multiple studies have shown that classical exponential smoothing models perform competitively, particularly when data are aggregated at monthly or quarterly frequency, but ARIMA models constantly obtain slightly better statistical fit.

3. Methodology

3.1 Software

The R statistical software environment, which offers comprehensive support for time series analysis and reproducibility, was used for all analyses.

3.2 Packages

In this analysis, the following packages are used:

- `forecast`: for implementing classical time series models such as Holt-Winters exponential smoothing and ARIMA-based modelling.
- `tseries`: for stationarity testing, such as KPSS and Augmented Dickey-Fuller (ADF) tests.
- `ASTSA`: for correlogram visualization and autocorrelation.
- `zoo`, `lubridate`, and `dplyr` - for data cleaning, manipulation, and aggregation of the high-frequency household electricity consumption data.
- `ggplot2`: to generate plots and displaying forecasts, trends, and seasonality.

3.3 Data Cleaning and Preparation

The original dataset contained missing values represented by question marks. These missing values were converted to NA during import. Missing observations in the `Global_active_power` variable were estimated using linear interpolation `na.approx()`

The data were aggregated from minute-level observations to monthly averages after interpolation. produced a final dataset spanning 48 full months.

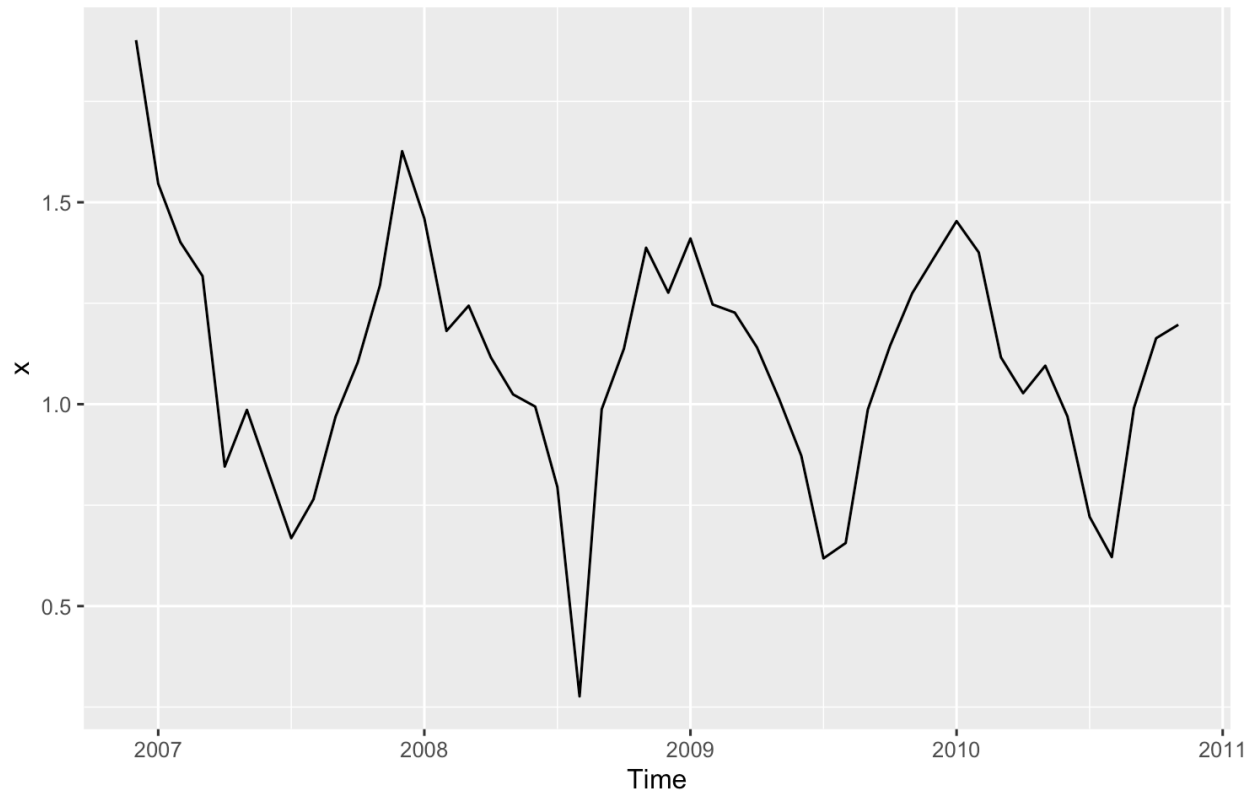
3.4 Preliminary Descriptive Analysis

Summary statistics were computed to examine the central tendency and dispersion of the series. The distribution of the data appeared symmetric, with no extreme outliers.

Describe and Shapiro test are used to confirm normal distribution from the data.

3.5 Time Series

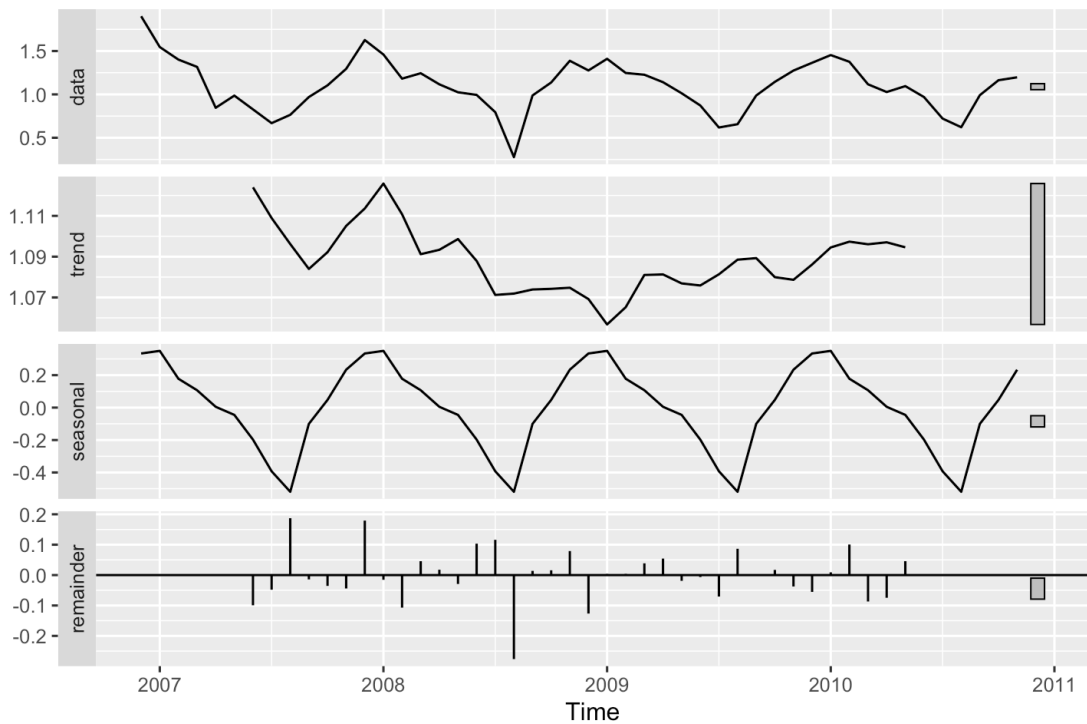
The monthly averages were converted into a time series dataset with a frequency of 12, representing monthly observations. The time series spans from December 2006 to November 2010.



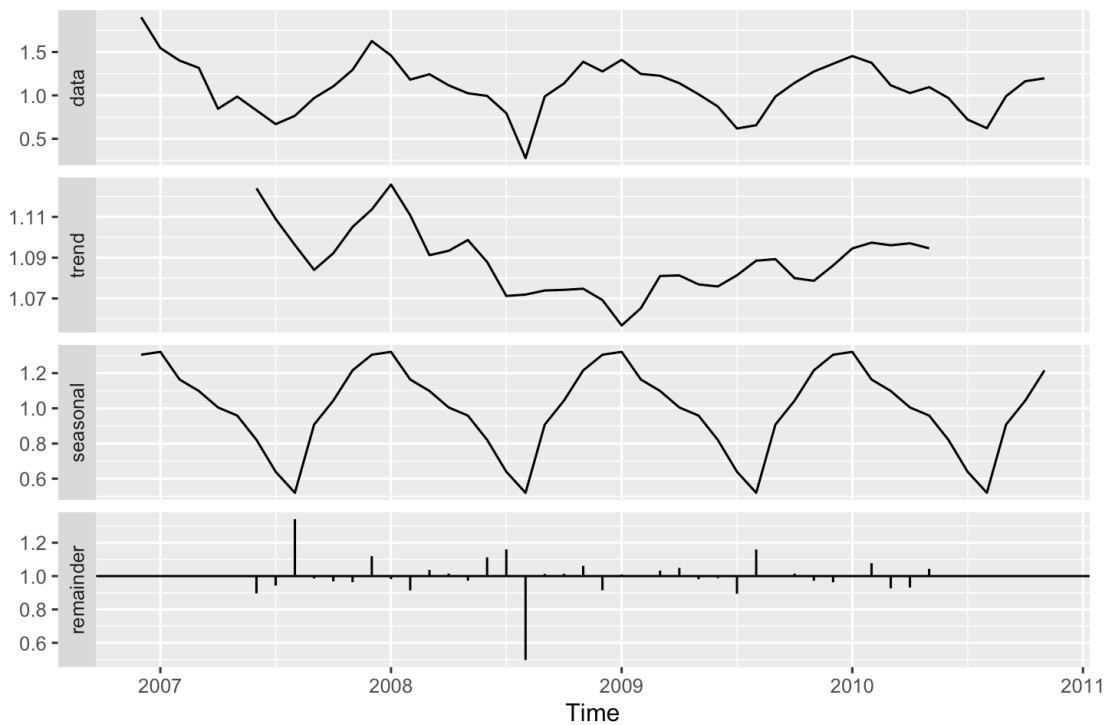
Visual inspection suggests a stable variance over time and clear seasonal fluctuations, indicating that an additive modelling approach may be appropriate.

- Seasonal decomposition was performed using both additive and multiplicative models.

Decomposition of additive time series



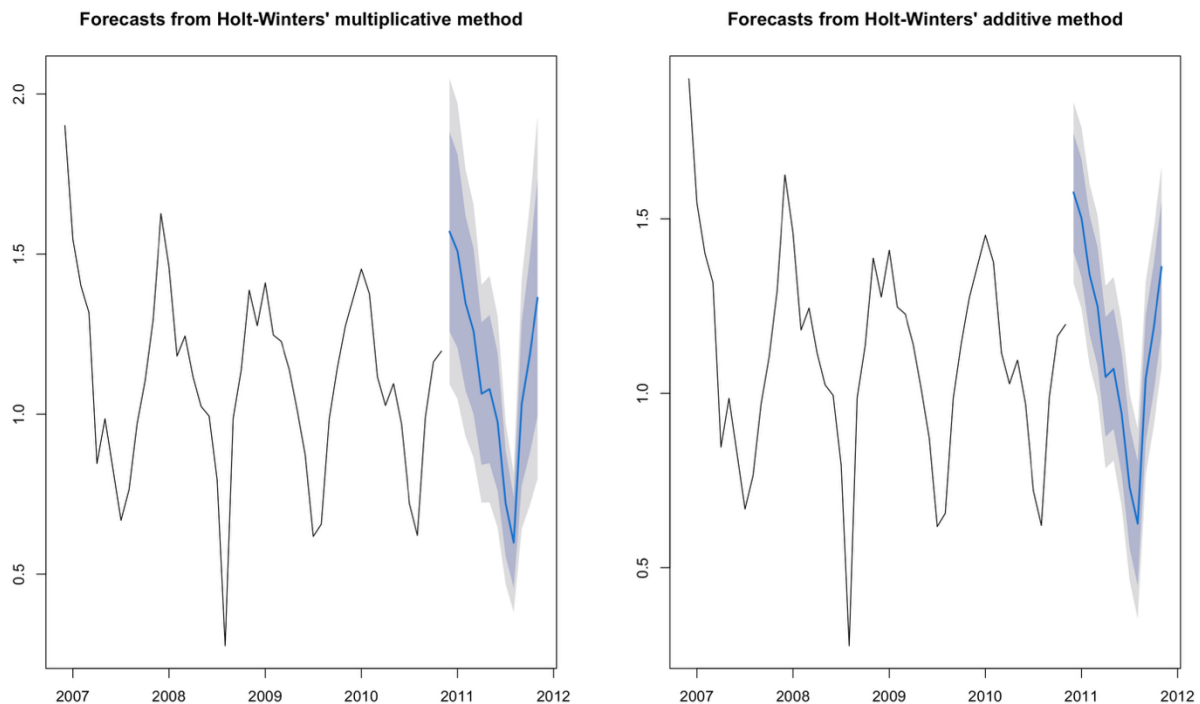
Decomposition of multiplicative time series



The additive decomposition showed consistent seasonality across years, while compared with the multiplicative decomposition and scale is also very low. This confirmed that an additive model is more suitable for this dataset.

3.6 Classical Time Series Modelling

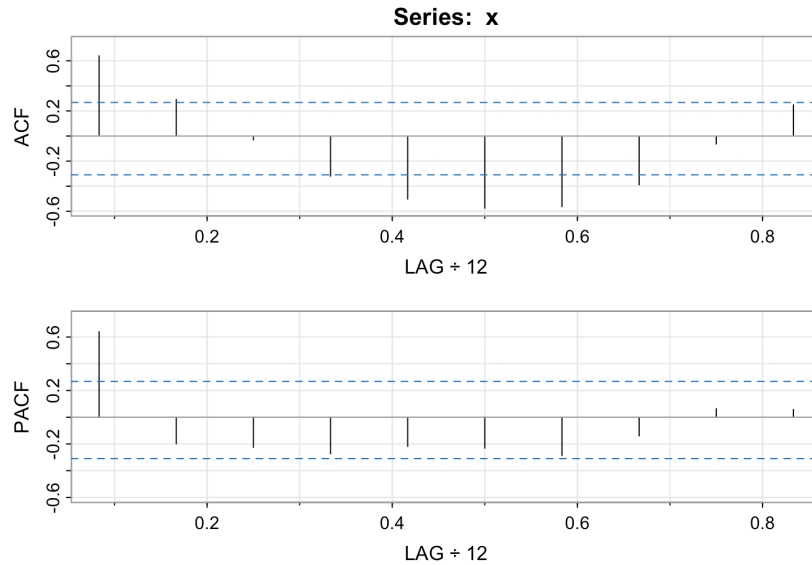
Holt–Winters exponential smoothing models were implemented with both additive and multiplicative seasonality.



Accuracy metrics such as MAE and RMSE showed no substantial differences between the two models. Additive model is preferred due to consistency with the decomposition analysis and stable variance.

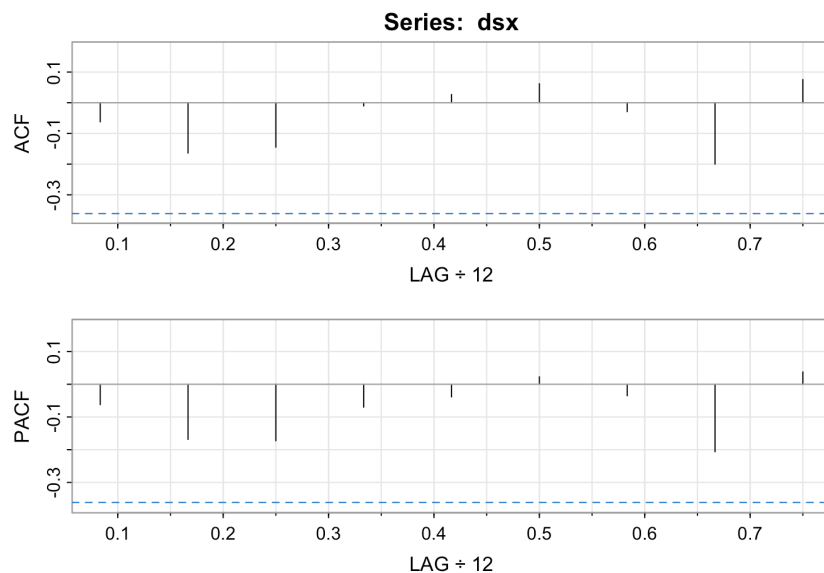
3.7 Stationarity Test

Stationarity is assessed using both correlogram and formal test statistics.



The Augmented Dickey-Fuller test failed and KPSS test indicates Stationarity. But seasonality is observed.

Therefore, Seasonal differencing with a lag of 12 was applied. And seasonality is removed confirmed with correlograms.



3.8 ARIMA Modelling

After seasonal differencing, autocorrelation analysis suggested a seasonal moving average component. Several SARIMA models were estimated and compared using AIC and residual diagnostics.

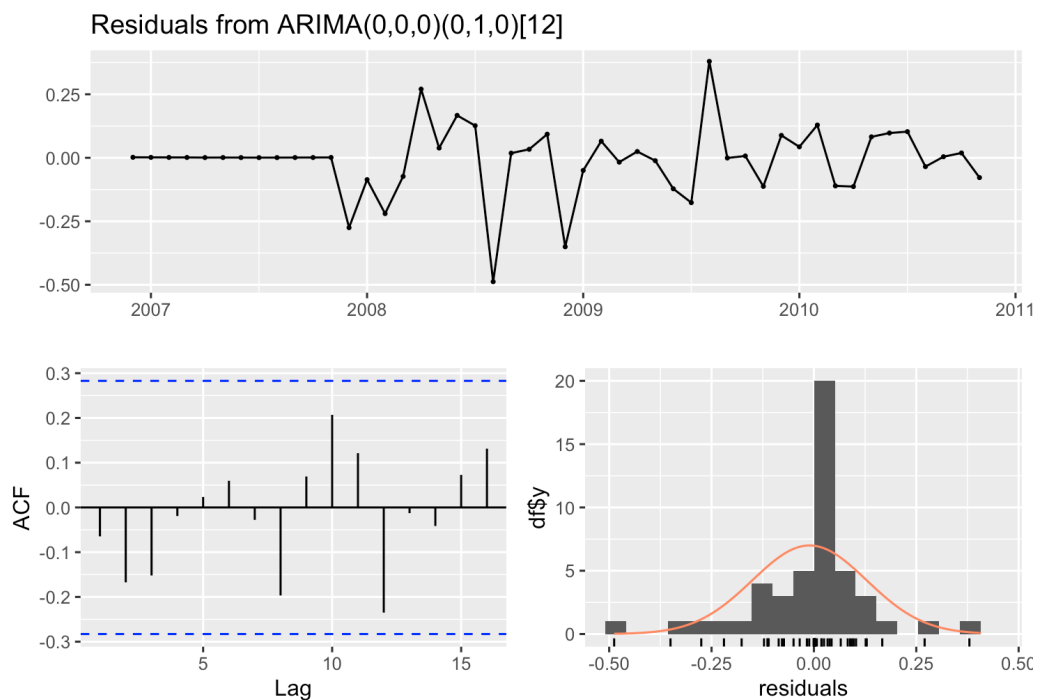
The final selected model was:

```
sarima(x,0,0,0,0,1,1,12)
```

Auto arima suggested:

```
sarima(x,0,0,0,0,1,0,12)
```

Since the parameter are less and AIC difference is not much choosing Auto Arima's suggested model.



Residuals exhibited no significant autocorrelation and followed an approximately normal distribution.

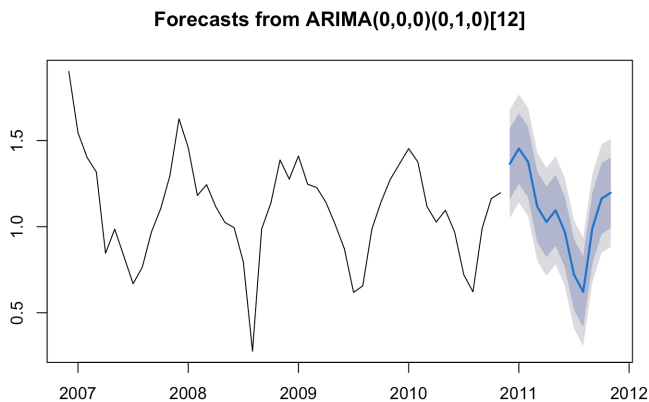
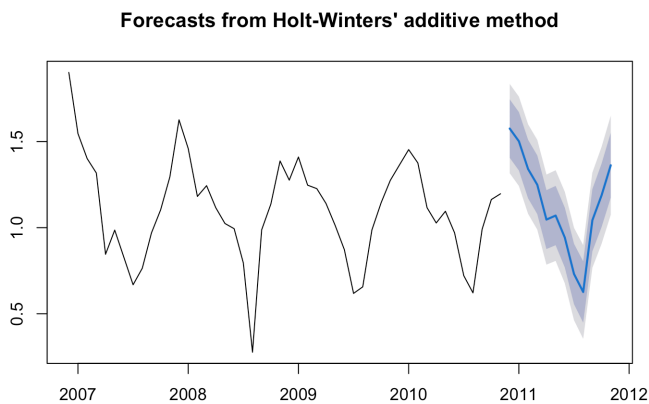
4. Results

4.1 Decomposition Results

Seasonality was identified as the most influential component of this time series, reflecting higher electricity consumption during colder months.

4.2 Forecasting Results

Twelve-month forecasts were generated using both the Holt-Winters additive model and the SARIMA model.



Both models produced similar forecasts, with seasonal patterns repeating consistently.

4.3 Model Comparison

While the SARIMA model achieved slightly better statistical fit, the Holt-Winters model offered greater interpretability and comparable predictive accuracy. Both models demonstrated strong forecasting performance.

To compare both Classical and Arima model:

- Train (till November 2009) and Test dataset are created to check with observed values.
- Summary table between Actual, Classical and Arima is created for comparison

Month	Actual	HoltWinters	AutoARIMA
Dec 2009	1.36	1.51	1.28
Jan 2010	1.45	1.40	1.41
Feb 2010	1.38	1.21	1.25
Mar 2010	1.12	1.20	1.23
Apr 2010	1.03	0.97	1.14
May 2010	1.10	0.94	1.01
Jun 2010	0.97	0.84	0.87
Jul 2010	0.72	0.63	0.62
Aug 2010	0.62	0.51	0.66
Sep 2010	0.99	0.91	0.99
Oct 2010	1.16	1.06	1.14
Nov 2010	1.20	1.26	1.27

- Auto Arima prediction are closer to the actual value for most of the months
- RMSE is calculated for Classical and Arima model, and concluded Arima model is a much better statistical fit and better explains the time series when compared with Classical Models

5. Conclusions and Discussion

This analysis used both classical and ARIMA models to analyze residential electricity use. The results show that the time series has a comparatively steady variance and high annual seasonality, which supports the application of additive model. At a 12-month frequency, seasonal trends were easily identified. Both the Auto ARIMA (seasonal ARIMA) model and the Holt-Winters additive model generated accurate and dependable forecasts.

The Auto ARIMA model generated fewer forecast errors, indicating higher predictive ability, according to an out-of-sample evaluation utilizing the last 12 months of observed data. This implies that the data's autocorrelation and seasonal structure are both successfully captured by Auto ARIMA.

The Auto ARIMA model offers a more reliable and statistically robust for predicting household power consumption, even though the Holt-Winters model remains valuable for practical applications because of its simplicity and ease of understanding.